

Maximum Likelihood Parameter Estimation and Bayes Decision Boundaries

Lab Assignment 5 Report

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1 Introduction and Objectives

In this assignment, we are using Maximum Likelihood Estimation (MLE) for parameter estimation for a 2d gaussian distributions from generated data. We then classify the test data using the Bayes decision rule and analyze how the different covariance matrix structures affect the shape and orientation of decision boundaries across all three classes.

The primary objectives are:

1. Generate 10,000 2-D Gaussian data points for each of three classes $\mathcal{N}(\mu_i, \Sigma_i)$ ($i = 1, 2, 3$).
2. Implement and evaluate six different covariance configurations:
 - Identical covariance across classes: (a) Isotropic ($\sigma^2 I$), (b) Diagonal, and (c) Full covariance.
 - Different covariances across classes: (a) Isotropic, (b) Diagonal, and (c) Full covariance.
3. Split the dataset into 80% training data (24,000 samples) and 20% test data (6,000 samples).
4. Estimate the class parameters ($\hat{\mu}_i, \hat{\Sigma}_i, \hat{P}(\omega_i)$) from the training data using Maximum Likelihood Estimation.
5. Then classify the test data using the Bayes decision rule, compute test accuracy and confusion matrices, and plot the decision boundaries with the test data overlaid.

2 Experiments and Results

The true class centers are kept constant across all experiments:

$$\mu_1 = \begin{bmatrix} -2.0 \\ 0.0 \end{bmatrix}, \quad \mu_2 = \begin{bmatrix} 2.0 \\ 0.0 \end{bmatrix}, \quad \mu_3 = \begin{bmatrix} 0.0 \\ 3.0 \end{bmatrix} \quad (1)$$

For each class, 10,000 samples were generated (30,000 samples total). In every case, 80% (24,000 samples) are used for training and 20% (6,000 samples) are used for testing.

2.1 Identical Covariance Across Classes

In Cases 1, 2, and 3, all three classes have the same covariance matrix. As shown in Figure 1, all decision boundaries are piecewise straight lines.

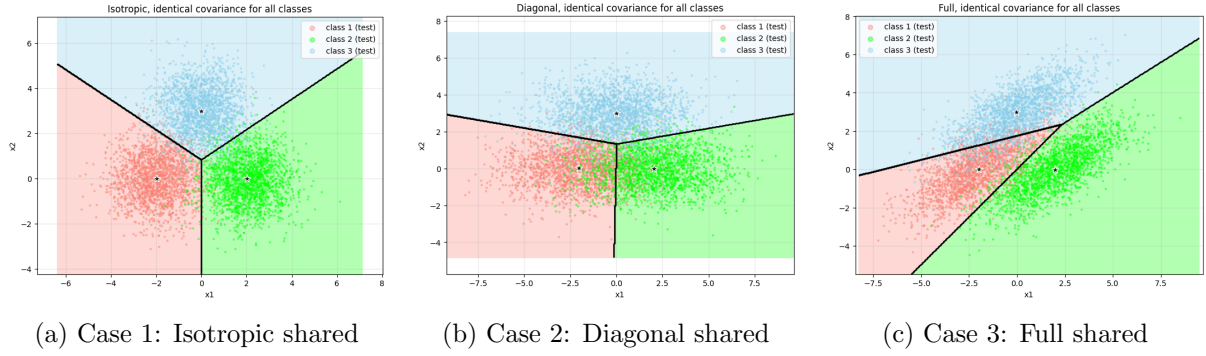


Figure 1: Bayes decision boundaries and test data for identical covariance cases.

2.1.1 Case 1: Isotropic, Identical Covariance ($\Sigma = \mathbf{I}$)

- **Target Covariance:** $\Sigma_1 = \Sigma_2 = \Sigma_3 = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$.
- **Estimated Parameters:** $\hat{\mu}_1 = [-1.99, -0.00]^T$, $\hat{\mu}_2 = [2.00, -0.01]^T$, $\hat{\mu}_3 = [-0.01, 2.99]^T$.
Common $\hat{\Sigma} = \begin{bmatrix} 0.999 & 0.000 \\ 0.000 & 0.999 \end{bmatrix}$. Estimated priors $\hat{P} = [0.333, 0.333, 0.333]$.
- **Test Accuracy:** 94.55%.

Observations:

- Here the decision boundaries are straight lines which acts as perpendicular bisectors between each pair of class means.
- Since all classes have the same variance ($\sigma^2 = 1.0$), the points are classified based on the nearest class center using Euclidean distance.
- As shown in the Table 1, misclassification is very low because the clusters are clearly separated.

True Class	Pred Class 1	Pred Class 2	Pred Class 3
Class 1	1907	37	56
Class 2	36	1916	48
Class 3	63	75	1862

Table 1: Confusion matrix for Case 1 (Test Accuracy: 94.55%).

2.1.2 Case 2: Diagonal, Identical Covariance ($\Sigma = \text{diag}(4, 1)$)

- **Target Covariance:** $\Sigma_1 = \Sigma_2 = \Sigma_3 = \begin{bmatrix} 4.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$.
- **Estimated Parameters:** $\hat{\mu}_1 = [-2.04, 0.02]^T$, $\hat{\mu}_2 = [2.02, -0.00]^T$, $\hat{\mu}_3 = [-0.00, 2.98]^T$.
Common $\hat{\Sigma} = \begin{bmatrix} 4.025 & 0.000 \\ 0.000 & 1.014 \end{bmatrix}$.
- **Test Accuracy:** 84.70%.

Observations:

- The decision boundary between Class 1 and Class 2 remains as a vertical line at $x_1 \approx 0$, while the boundaries that separate Class 3 are slanted straight lines.

- The large horizontal variance ($\sigma_{x_1}^2 = 4.0$) causes Class 1 and Class 2 to stretch toward each other.
- As there is overlapping, it reduces the accuracy to 84.70%, with significant misclassifications between Class 1 and Class 2 (Table 2).

True Class	Pred Class 1	Pred Class 2	Pred Class 3
Class 1	1609	314	77
Class 2	300	1606	94
Class 3	99	70	1831

Table 2: Confusion matrix for Case 2 (Test Accuracy: 84.70%).

2.1.3 Case 3: Full, Identical Covariance ($\Sigma = \begin{bmatrix} 3.0 & 1.5 \\ 1.5 & 1.5 \end{bmatrix}$)

- **Target Covariance:** $\Sigma_1 = \Sigma_2 = \Sigma_3 = \begin{bmatrix} 3.0 & 1.5 \\ 1.5 & 1.5 \end{bmatrix}$.
- **Estimated Parameters:** $\hat{\mu}_1 = [-2.00, 0.01]^T$, $\hat{\mu}_2 = [1.98, -0.02]^T$, $\hat{\mu}_3 = [-0.02, 2.99]^T$.
Common $\hat{\Sigma} = \begin{bmatrix} 2.983 & 1.502 \\ 1.502 & 1.509 \end{bmatrix}$.
- **Test Accuracy:** 89.77%.

Observations:

- The positive covariance ($\sigma_{12} = 1.5$) tilts the data clusters along the diagonal direction.
- The decision boundaries remain as straight lines.
- From the confusion matrix, diagonal tilt clearly separates Class 2 and Class 3 (only 4 misclassifications), while most errors occur between Class 1 and Class 3 along the direction of the elongation (Table 3).

True Class	Pred Class 1	Pred Class 2	Pred Class 3
Class 1	1732	89	179
Class 2	113	1883	4
Class 3	197	8	1795

Table 3: Confusion matrix for Case 3 (Test Accuracy: 89.77%).

2.2 Different Covariances Across Classes

In Cases 4, 5, and 6, each class has a distinct covariance matrix. Because the quadratic terms $x^T \hat{\Sigma}_k^{-1} x$ do not cancel, all decision boundaries become curved quadratic contours (Figure 2).

2.2.1 Case 4: Isotropic, Different Covariances ($\sigma_1^2 = 1.0, \sigma_2^2 = 2.0, \sigma_3^2 = 0.5$)

- **Target Covariances:** $\Sigma_1 = \mathbf{I}$, $\Sigma_2 = 2\mathbf{I}$, $\Sigma_3 = 0.5\mathbf{I}$.
- **Estimated Parameters:** $\hat{\mu}_1 = [-2.00, -0.01]^T$, $\hat{\mu}_2 = [1.99, 0.02]^T$, $\hat{\mu}_3 = [-0.01, 3.00]^T$.
 $\hat{\Sigma}_1 \approx \begin{bmatrix} 1.011 & 0 \\ 0 & 0.970 \end{bmatrix}$, $\hat{\Sigma}_2 \approx \begin{bmatrix} 2.051 & 0 \\ 0 & 1.998 \end{bmatrix}$, $\hat{\Sigma}_3 \approx \begin{bmatrix} 0.511 & 0 \\ 0 & 0.508 \end{bmatrix}$.

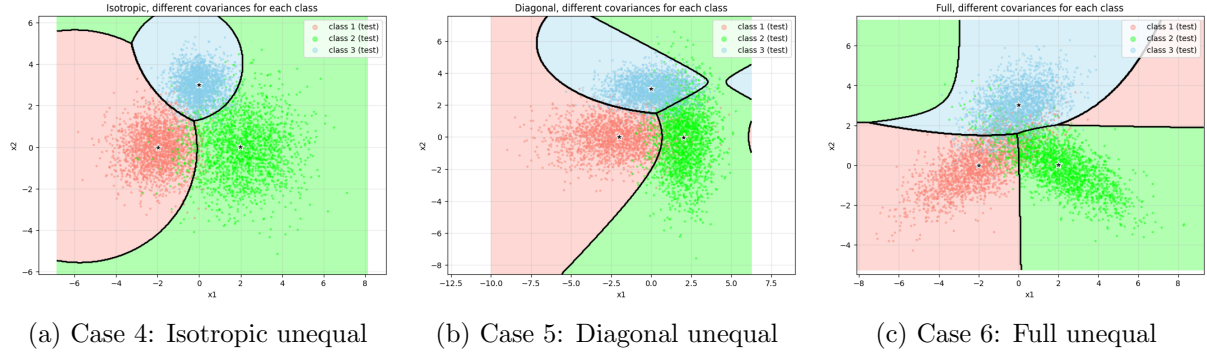


Figure 2: Bayes decision boundaries and test data for different covariance cases.

- **Test Accuracy:** 93.95%.

Observations:

- The decision boundaries are enclosed around the classes with smaller variance.
- Class 3 has the smallest variance ($\sigma_3^2 = 0.5$), thus it is enclosed by the boundary. It achieves the highest accuracy ($> 97\%$).
- Class 2 has the largest variance ($\sigma_2^2 = 2.0$), so its decision region expands and encloses the other classes at larger distances (Table 4).

True Class	Pred Class 1	Pred Class 2	Pred Class 3
Class 1	1902	66	32
Class 2	100	1810	90
Class 3	17	42	1941

Table 4: Confusion matrix for Case 4 (Test Accuracy: 93.95%).

2.2.2 Case 5: Diagonal, Different Covariances

- **Target Covariances:** $\Sigma_1 = \text{diag}(4, 1)$, $\Sigma_2 = \text{diag}(1, 4)$, $\Sigma_3 = \text{diag}(2, 0.5)$.
- **Estimated Parameters:** $\hat{\mu}_1 \approx [-1.99, -0.00]^T$, $\hat{\mu}_2 \approx [2.02, -0.03]^T$, $\hat{\mu}_3 \approx [-0.01, 3.00]^T$.
 $\hat{\Sigma}_1 \approx \text{diag}(4.01, 1.02)$, $\hat{\Sigma}_2 \approx \text{diag}(1.01, 4.01)$, $\hat{\Sigma}_3 \approx \text{diag}(2.00, 0.50)$.
- **Test Accuracy:** 88.55%.

Observations:

- Class 1 is stretched horizontally, Class 2 is stretched vertically, and Class 3 has moderate horizontal spread.
- The decision boundaries are curved quadratic conic sections (hyperbolic and parabolic segments).
- There is a moderate overlap between class 1 and class 2 (Table 5).

2.2.3 Case 6: Full, Different Covariances

- **Target Covariances:** $\Sigma_1 = \begin{bmatrix} 3.0 & 1.5 \\ 1.5 & 1.5 \end{bmatrix}$, $\Sigma_2 = \begin{bmatrix} 3.0 & -1.5 \\ -1.5 & 1.5 \end{bmatrix}$, $\Sigma_3 = \begin{bmatrix} 1.5 & 0.5 \\ 0.5 & 1.0 \end{bmatrix}$.

True Class	Pred Class 1	Pred Class 2	Pred Class 3
Class 1	1746	192	62
Class 2	90	1680	230
Class 3	42	89	1869

Table 5: Confusion matrix for Case 5 (Test Accuracy: 88.55%).

- **Estimated Parameters:** $\hat{\mu}_1 \approx [-2.00, -0.00]^T$, $\hat{\mu}_2 \approx [1.99, 0.01]^T$, $\hat{\mu}_3 \approx [-0.01, 3.02]^T$.
 $\hat{\Sigma}_1 \approx \begin{bmatrix} 3.02 & 1.48 \\ 1.48 & 1.50 \end{bmatrix}$, $\hat{\Sigma}_2 \approx \begin{bmatrix} 2.97 & -1.51 \\ -1.51 & 1.52 \end{bmatrix}$, $\hat{\Sigma}_3 \approx \begin{bmatrix} 1.50 & 0.50 \\ 0.50 & 1.00 \end{bmatrix}$.
- **Test Accuracy:** 86.70%.

Observations:

- Class 1 is tilted upward (positive covariance), while Class 2 is tilted downward (negative covariance).
- This difference creates curved and complex decision boundaries.
- Misclassifications are distributed across all three classes because of overlapping (Table 6).

True Class	Pred Class 1	Pred Class 2	Pred Class 3
Class 1	1655	164	181
Class 2	137	1706	157
Class 3	111	59	1830

Table 6: Confusion matrix for Case 6 (Test Accuracy: 86.70%).

2.3 Summary of Results Across All Cases

Table 7 summarizes the performance and boundary types for all six cases.

Case	Covariance Structure	Covariance Equality	Boundary Shape	Test Accuracy
1	Isotropic ($\sigma^2 I$)	Identical ($\Sigma_i = \Sigma$)	Linear (straight lines)	94.55%
2	Diagonal ($\text{diag}(\sigma_1^2, \sigma_2^2)$)	Identical ($\Sigma_i = \Sigma$)	Linear (slanted lines)	84.70%
3	Full (Σ)	Identical ($\Sigma_i = \Sigma$)	Linear (rotated lines)	89.77%
4	Isotropic ($\sigma_i^2 I$)	Different ($\Sigma_i \neq \Sigma_j$)	Quadratic (circular arcs)	93.95%
5	Diagonal ($\text{diag}(\sigma_{i1}^2, \sigma_{i2}^2)$)	Different ($\Sigma_i \neq \Sigma_j$)	Quadratic (axis-aligned curves)	88.55%
6	Full (Σ_i)	Different ($\Sigma_i \neq \Sigma_j$)	Quadratic (rotated curves)	86.70%

Table 7: Comparison of classification results across the six covariance configurations.

3 Conclusion

In this assignment, we generated 2-D Gaussian synthetic data for three classes under six different covariance configurations and performed Maximum Likelihood parameter estimation using an 80% training split. We then classified the test data using the Bayes decision rule and checked the accuracy and confusion matrix.